



نظرية الحسابات

Theory of Computation

203 Comp

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توزيع الدرجات :

70 درجة

نظري فاي نهاية الفصل

25 درجة

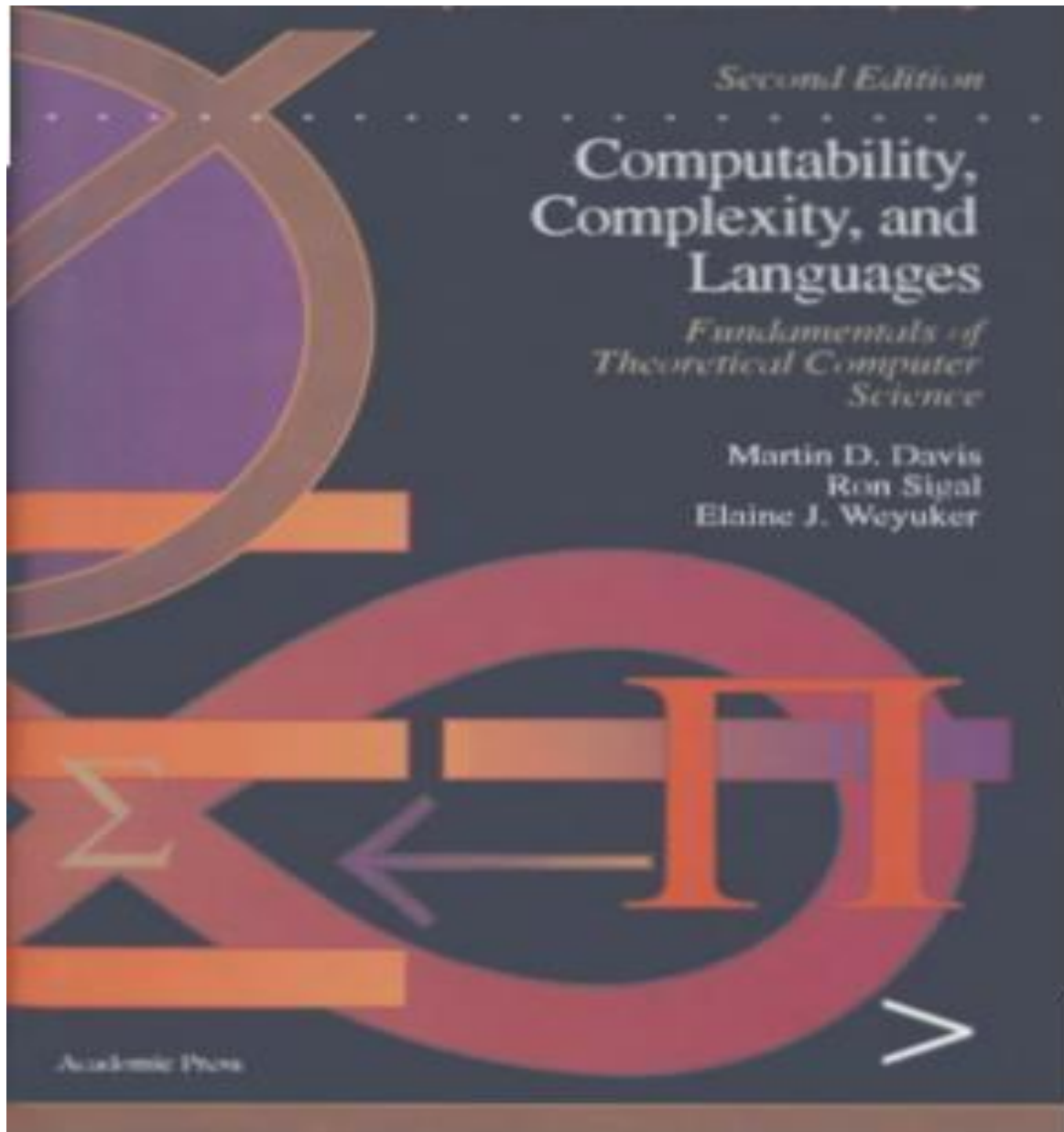
أعمال فصلية

5 درجة

شفوي

100 درجة

المجموع





Chapter I

Preliminaries

Chapter Contents:

1. Sets and n -tuples
2. Functions
3. Alphabets and Strings
4. Predicates
5. Quantifiers
6. Proof by Contradiction
7. Mathematical Induction

1. Sets and n -tuples



Sets

Definition : A set is an unordered collection of distinct objects.

Members of a Set

Definition: The objects in a set are called elements or members of a set. A set is said to contain its elements

Notation, for a set A:

$x \in A$: x is an element of A

$x \notin A$: x is not an element of A

Sets can be denoted using **uppercase letters**.

Lowercase letters are used to denote **elements of sets**.



Null set or Empty set

- A set that has no elements is called the **empty set** or **null set** and is denoted $\emptyset$
- A set that has one element is called a **singleton set**.
 - For example: $\{a\}$, with brackets, is a singleton set
 - a , without brackets, is an element of the set $\{a\}$
- Note $\emptyset \neq \{\emptyset\}$
 - The left-hand side is the empty set
 - The right hand-side is a singleton set, and a set containing a set

Subset of a Set

- **Definition:** A is said to be a subset of B , and we write $A \subseteq B$, if and only if every element of A is also an element of B
- That is, we have the equivalence:

$$A \subseteq B \Leftrightarrow \forall x(x \in A \Rightarrow x \in B)$$

- Example:-

$$A = \{1, 2, 3, 4\} \quad B = \{1, 3, 4\} \quad C = \{4, 5, 6\}$$

$$B \subseteq A \rightarrow B \text{ is a Subset of } A$$

$$C \subseteq A \rightarrow C \text{ is not a subset of } A$$

- Example:-

$$A = \{2, 4, 17, 23\} \quad B = \{2, 4, 17, 23\}$$

$$A \subseteq B$$

$$B \subseteq A$$

$$A = B$$

- Thus, for any set A , it is true that $A \subseteq A$
- Also, for any set A , it is true that $\emptyset \subseteq A$

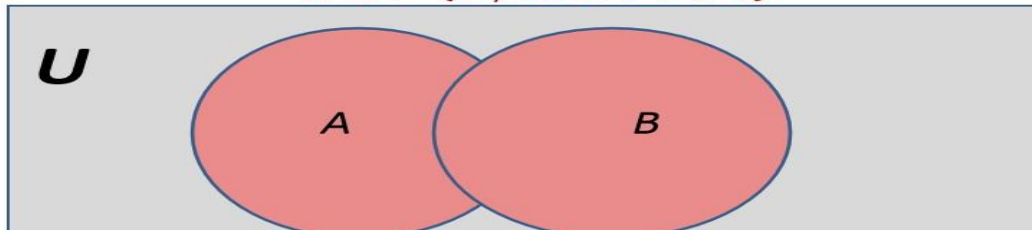


Set Operations

1. Union

- Definition :** Let A and B be sets. The union of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B , or in both
- An element x belongs to the union of the sets A and B if and only if x belongs to A or x belongs to B . This tells us that.

$$A \cup B = \{x / x \in A \vee x \in B\}$$

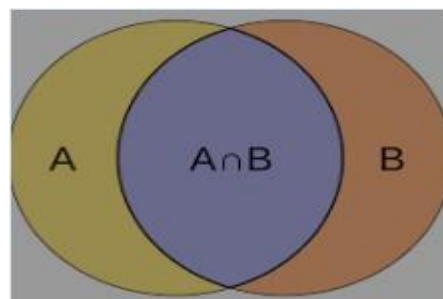


Set Operations

2. Intersection

- Definition :** Let A and B be sets. The intersection of the sets A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .
- An element x belongs to the intersection of the sets A and B if and only if x belongs to A and x belongs to B . This tells us that

$$A \cap B = \{x / x \in A \wedge x \in B\}$$



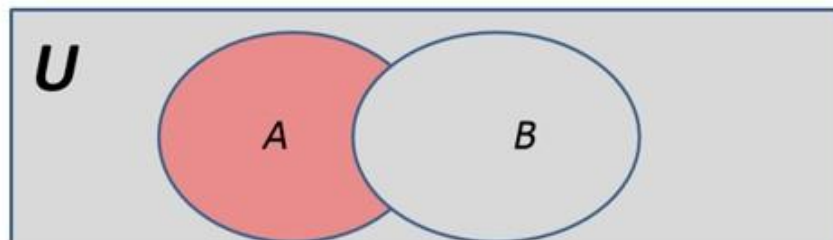


Set Operations

3. Set Difference

- **Definition:** Let A and B be sets. The difference of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . The difference of A and B is also called the complement of B with respect to A .
- An element x belongs to the difference of A and B if and only if $x \in A$ and $x \notin B$. This tells us that

$$A - B = \{x / x \in A \wedge x \notin B\}.$$



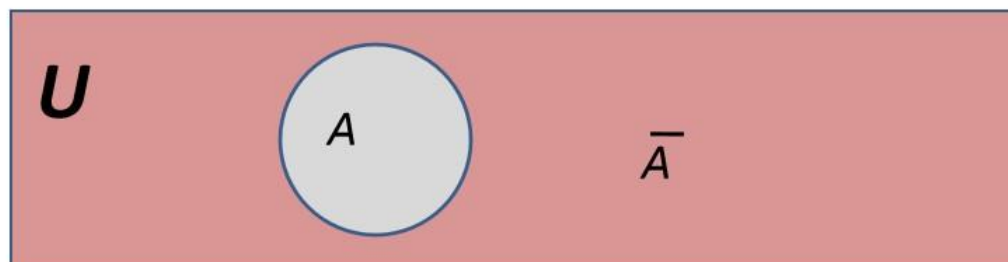
$$A - B = A - (A \cap B)$$

Set Operations

4. Set Complement

- **Definition:** Let U be the universal set. The complement of the set A , denoted by $\bar{A}$, is the complement of A with respect to U . Therefore, the complement of the set A is $U - A$.
- An element belongs to $\bar{A}$ if and only if $x \notin A$. This tells us that

$$\bar{A} = A^c = \{x \in U / x \notin A\}.$$





5. The De Morgan identities

$$\overline{R \cup S} = \bar{R} \cap \bar{S},$$

$$\overline{R \cap S} = \bar{R} \cup \bar{S}$$

Cartesian Products

- The order of elements in a collection is often important. Because sets are unordered, a different structure is needed to represent ordered collections. This is provided by **ordered n -tuples**.

Definition: The **ordered n -tuple** $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, $\dots$, and a_n as its n^{th} element.

- Two ordered n -tuples $(a_1, a_2, \dots, a_n)$ and $(b_1, b_2, \dots, b_n)$ are equal if and only if for every $i=1, 2, \dots, n$ we have $a_i=b_i$ $(a_1, a_2, \dots, a_n)$
- A 2-tuple ($n=2$) is called an **ordered pair**
- Thus the ordered pair $(3, 5)$ is *different* to $(5, 3)$
- Note the use of *parentheses* ('round brackets'), and not braces ('curly brackets') as for sets
- So $(1, 4, 5) \neq (1, 5, 4)$ (but $\{1, 4, 5\} = \{1, 5, 4\}$)



Cartesian Product

Definition: Let A and B be two sets. The Cartesian product (Cross Product) of A and B , denoted $A \times B$, is the set of all ordered pairs (a,b) where $a \in A$ and $b \in B$

$$A \times B = \{ (a,b) \mid (a \in A) \wedge (b \in B) \}$$

Example: If A is the set of digits 0-9, & B is the set of letters a - z , then $(3, t)$ is in $A \times B$, but $(m, 7)$ is *not* in $A \times B$ – although it *is* in $B \times A$

Example: If $A = \{1, 2, 3\}$ & $B = \{p, q\}$, then

$$\begin{aligned} A \times B &= \{(x, y) : x \in A \text{ and } y \in B\} \\ &= \{(1, p), (1, q), (2, p), (2, q), (3, p), (3, q)\} \end{aligned}$$

Example: $K = \{a, b, c\}$ and $L = \{1, 2\}$, then

- $K \times L = \{(a,1), (a,2), (b,1), (b,2), (c,1), (c,2)\}$
- $L \times K = \{(1,a), (2,a), (1,b), (2,b), (1,c), (2,c)\}$
- $L \times L = \{(1,1), (1,2), (2,1), (2,2)\}$

$A \times B \neq B \times A$ unless $A = \emptyset$ or $B = \emptyset$ or $A = B$. Exercise : Find a example to prove this.



Cartesian Product of a Set with Itself

- A^2 denote $A \times A$, the **Cartesian product of the set A with itself**.
- Similarly, $A^3 = A \times A \times A$, $A^4 = A \times A \times A \times A$, and so on. More generally,
- $A^n = A \times A \times \dots \times A$ (n times) = $\{(a_1, a_2, \dots, a_n) \mid a_i \in A \text{ for } i = 1, 2, \dots, n\}$.

Example: Suppose that $A = \{1, 2\}$. Compute A^2 and A^3

$A^2 = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$ and

$A^3 = \{(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2), (2, 1, 1), (2, 1, 2), (2, 2, 1), (2, 2, 2)\}$.

Example: $\{0, 1\}^2 = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$

Example: The elements of $\{0, 1\}^n$ are **ordered n -tuples** in which each element is **0** or **1** – so a typical element of $\{0, 1\}^6$ is **(0, 1, 1, 1, 0, 1)**

- Think of $\{0, 1\}^n$ as the set of all strings of n bits
- **Note** that $L \times L \times L \times D \times D \times D = L^3 \times D^3$

Cartesian Product of n Sets

- Cartesian Products of more than two set can also be defined

Definition: The **Cartesian product of n sets**, $A_1, A_2, \dots, A_n$, denoted $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$ where a_i belongs to A_i for $i = 1, 2, \dots, n$

- i.e., $A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i=1, 2, \dots, n\}$

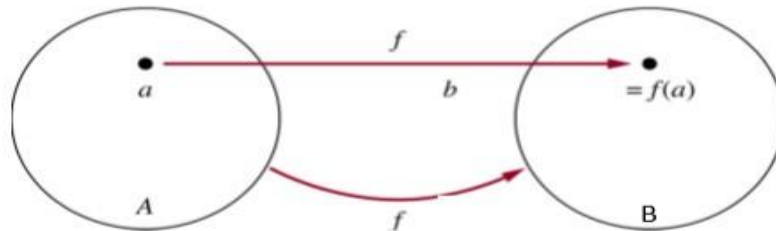
Example: A car reg. no. such as KCT454 can be regarded as an ordered 6-tuple (K, C, T, 4, 5, 4).

- If L is the set of all letters, & D is the set of all decimal digits, then the set of all possible car registration is $L \times L \times L \times D \times D \times D$



2. Functions

Function



Definition:

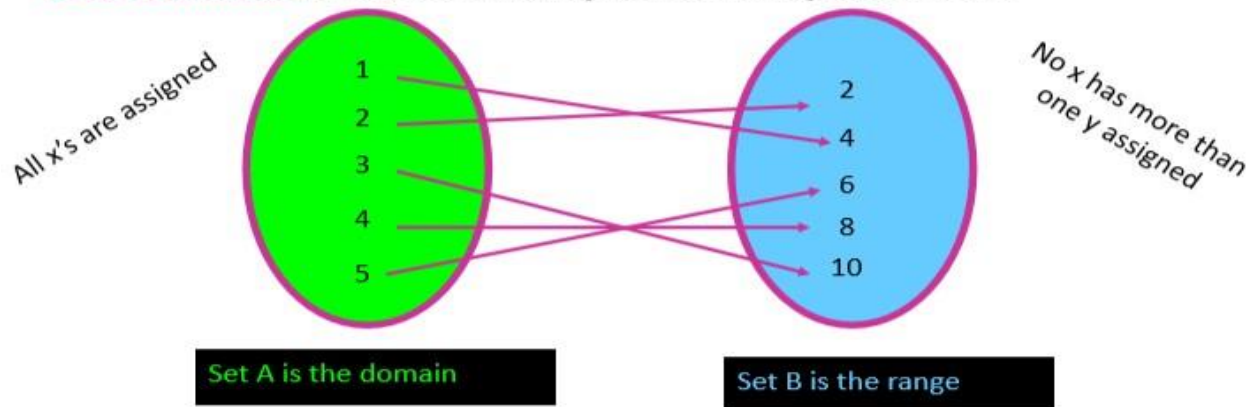
Given A and B , nonempty sets, a function f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a)=b$ if b is the element of B assigned by function f to the element a of A .

If f is a function from A to B , we write $f : A \rightarrow B$.

Note: Functions are also called **mappings** or **transformations**.

We can consider f as a set of pairs $f = \{(a, b)\}$

A **function** f from set A to set B is a rule of correspondence that assigns to **each element** x in the set A exactly one element y in the set B .



This is a function ---it meets our conditions

Must use all the x 's



The x value can only be assigned to one y

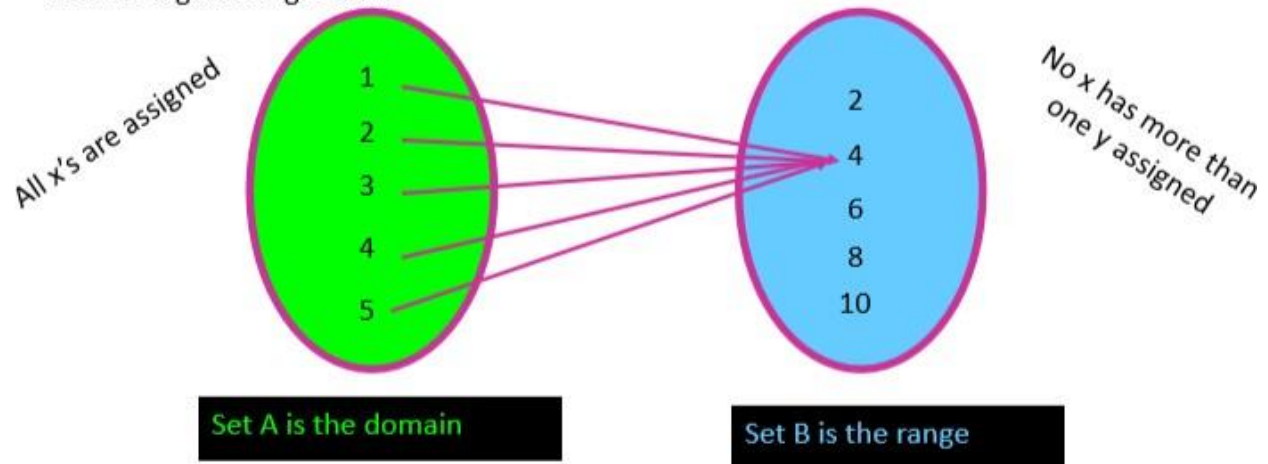


$(a, b) \in f$ and $(a, c) \in f$ implies $b = c$.



$$F = \{(1, 4), (2, 2), (3, 10), (4, 8), (5, 6)\}$$

The second condition says each x can have only one y, but it **CAN** be the same y as another x gets assigned to.

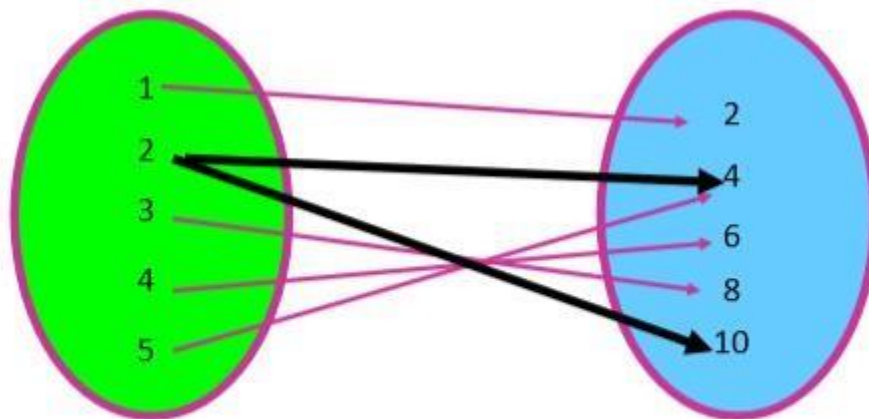


This is a function ---it meets our conditions

Must use all the x's



The x value can only be assigned to one y



2 was assigned both 4 and 10

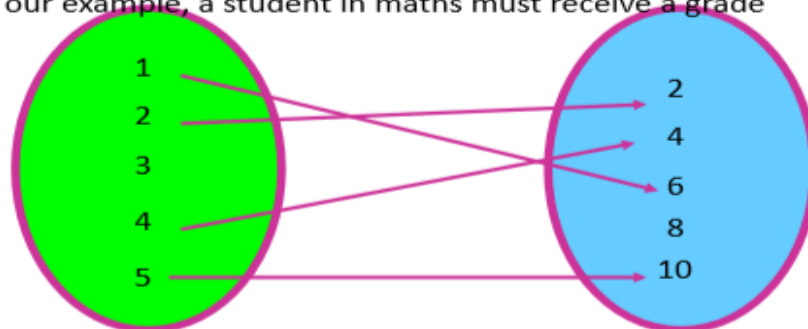
Is the relation shown above a function?



Check this relation out to determine if it is a function

It is not---3 didn't get assigned to anything

Comparing to our example, a student in maths must receive a grade



Set A is the domain

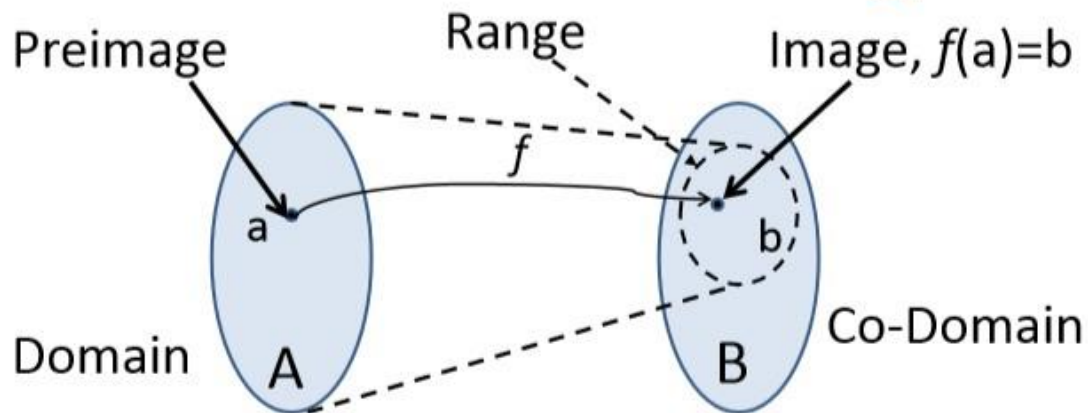
Set B is the range

**This is not a function---
it doesn't assign each
x with a y**

Must use all the x's

The x value can only be assigned to one y

Some Function Terminology



Let $f: A \rightarrow B$ and $f(a)=b$. Then we use the following terminology:

A is the **domain** of f , denoted **$\text{dom}(f)$**

B is the **co-domain** of f

b is the **image** of a

a is the **preimage (antecedent)** of b

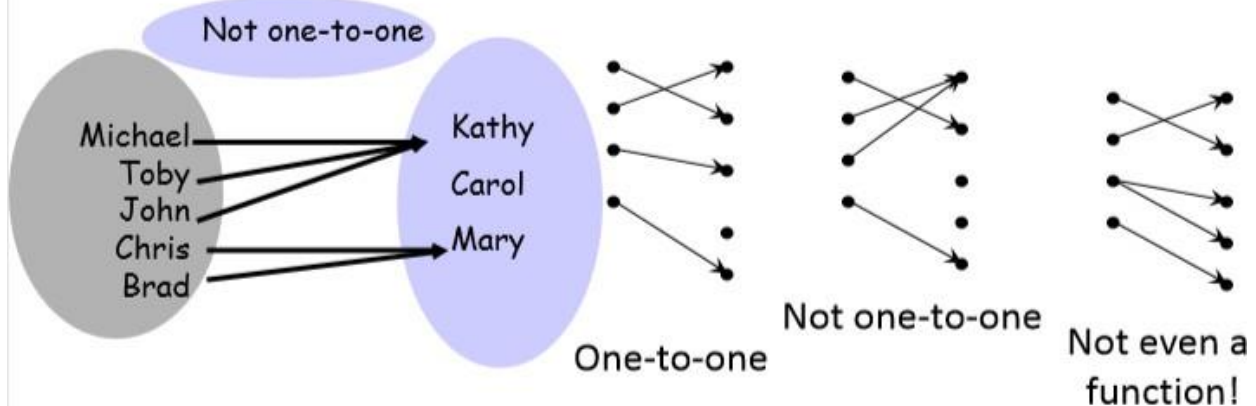
The **range** of f is the set of all images of elements of A, denoted **$\text{rng}(f)$**



Every $b \in B$ has at most 1 preimage.

Functions - Injection

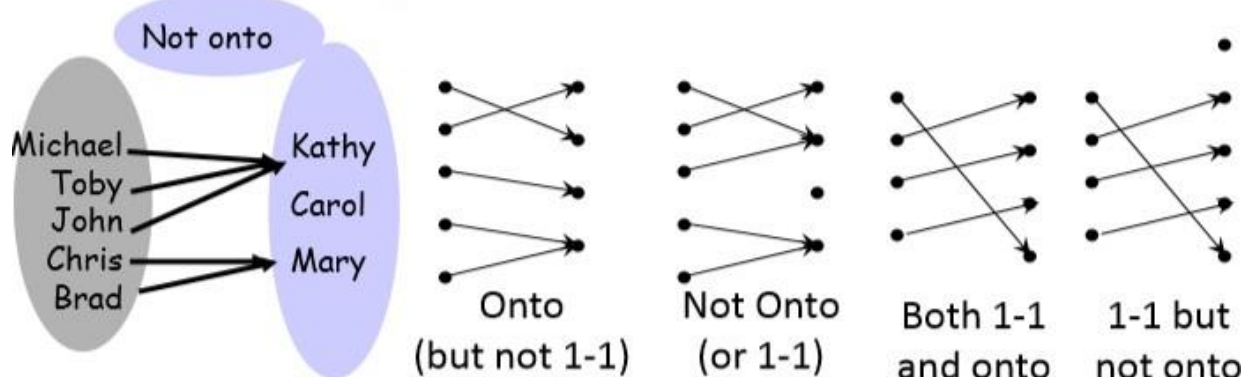
Definition : A function f is said to be one-to-one, or an injection, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be **injective if it is one-to-one**.

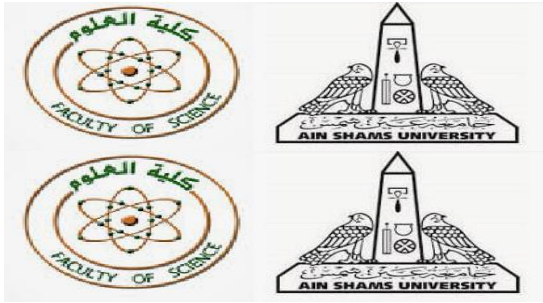


Every $b \in B$ has at least 1 preimage.

Functions - Surjection

Definition : A function f from A to B is called onto, or a surjection, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$. A function f is called **surjective if it is onto**.

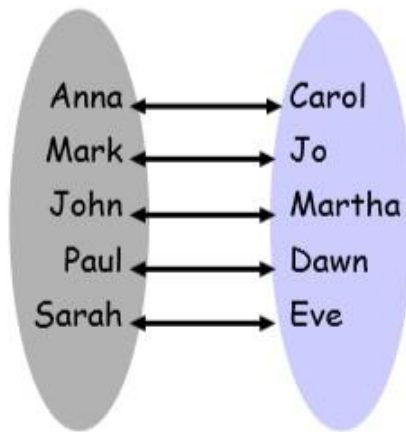




Every $b \in B$ has exactly 1 preimage.

Functions - Bijection

Definition : The function f is a **one-to-one correspondence, or a bijection**, if it is **both one-to-one and onto**. We also say that such a function is **bijective**.



3. Predicates.

- Predicate is a Boolean-valued function on a set S , such that for each $a \in S$, either

$$P(a) = \text{TRUE} \text{ or } P(a) = \text{FALSE},$$

- It is useful to identify the truth values with specific numbers, so we set

$$\text{TRUE} = 1 \text{ and } \text{FALSE} = 0.$$

- A predicate is a special kind of function with values in $\{0, 1\}$.
- Predicates on a set S are usually specified by expressions which

become statements, either true or false, when variables in the



expression are replaced by symbols designating fixed elements of S . or example, the expression

$$x < 5$$

specifies a predicate on N , namely,

$$P(x) = \begin{cases} 1 & \text{if } x = 0, 1, 2, 3, 4 \\ 0 & \text{Otherwise} \end{cases}$$

- Three basic operations on truth values are defined by the following tables:

p	$\sim p$				
0	1	p	q	$p \& q$	$p \vee q$
1	0	1	1	1	1
		0	1	0	1
		1	0	0	1
		0	0	0	0

- Given a predicate P on a set S , there is a corresponding subset R of S , namely, the set of all elements $a \in S$ for which $P(a) = 1$ we write

$$R = \{a \in S | P(a)\}.$$

Conversely, given a subset R of a given set S , the expression

$$x \in R$$

defines a predicate on S , namely, the predicate defined by

$$P(x) = \begin{cases} 1 & \text{if } x \in R \\ 0 & \text{if } x \notin R \end{cases}$$



- For example, the expression

$$x < 5$$

specifies a predicate on N , namely,

$$R = \{0, \dots, 4\}$$

$$R(x) = \begin{cases} 1 & \text{if } x \in R \\ 0 & \text{Otherwise} \end{cases}$$

- To indicate that two expressions containing variables define the same predicate we place the symbol $\Leftrightarrow$ between them. Thus,

$$x < 5 \Leftrightarrow x = 0 \vee x = 1 \vee x = 2 \vee x = 3 \vee x = 4.$$

4. Quantifiers

Universal quantifier: $\forall$

Existential quantifier: $\exists$

Definition : The *universal quantification* of $P(x)$ is the statement “ $P(x)$ for all values of x in the domain.”

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$.

Here $\forall$ is called the **universal quantifier**. We read $\forall x P(x)$ as “**for all $x P(x)$** ” or “**for every $x P(x)$** .” An element for which $P(x)$ is false is called a **counterexample** of $\forall x P(x)$.

TABLE 1 Quantifiers.

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .



Example 1: $U = \{\text{all real numbers}\}$, $P(x): x+1 > x$. What is the truth value of $\forall x P(x)$
(or)

Let $P(x)$ be the statement " $x + 1 > x$." What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?

Solution

Because $P(x)$ is true for all real numbers x , the quantification $\forall x P(x)$ is true.

Example 2 : $U = \{\text{all real numbers}\}$, $Q(x): x < 2$. What is the truth value of $\forall x Q(x)$
(or)

Let $Q(x)$ be the statement " $x < 2$." What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

Solution:

$Q(x)$ is not true for every real number x , because, for instance, $Q(3)$ is false. That is, $x = 3$ is a counterexample for the statement $\forall x Q(x)$. **Thus $\forall x Q(x)$ is false.**

- If the elements in the universe of discourse can be listed, then

$$U = \{x_1, x_2, \dots, x_n\}$$

$$\forall x P(x) \Leftrightarrow P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$$

This conjunction is true iff only if $P(x_1), P(x_2), \dots, P(x_n)$ are all true.

Example : What is the truth value of $\forall x P(x)$, where $P(x)$ is the statement " $x^2 < 10$ " and the domain consists of the positive integers not exceeding 3?
and the domain consists of the positive integers not exceeding 4?

Solution : $P(x): x^2 < 10$

$U = \{\text{positive integers not exceeding 3}\}$

$U = \{\text{positive integers not exceeding 4}\}$

P(1)	P(2)	P(3)	P(1) $\wedge$ P(2) $\wedge$ P(3)
$1^2 < 10$ $1 < 10$ TRUE	$2^2 < 10$ $4 < 10$ TRUE	$3^2 < 10$ $9 < 10$ TRUE	$T \wedge T \wedge T =$ T

P(1)	P(2)	P(3)	P(4)	P(1) $\wedge$ P(2) $\wedge$ P(3) $\wedge$ P(4)
$1^2 < 10$ $1 < 10$ TRUE	$2^2 < 10$ $4 < 10$ TRUE	$3^2 < 10$ $9 < 10$ TRUE	$4^2 < 10$ $16 < 10$ FALSE	$T \wedge T \wedge T \wedge F =$ F



Existential Quantifier Examples

Example 1: $U = \{\text{all real numbers}\}$, $P(x): x > 3$. What is the truth value of $\exists x P(x)$

(or)

Let $P(x)$ denote the statement " $x > 3$." What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

Solution

Because " $x > 3$ " is sometimes true—for instance, when $x = 4$ —the existential quantification of $P(x)$, which is $\exists x P(x)$, is true.

Example 2 : $U = \{\text{all real numbers}\}$, $Q(x): x = x + 1$. What is the truth value of $\exists x Q(x)$

(or)

Let $Q(x)$ denote the statement " $x = x + 1$." What is the truth value of the quantification $\exists x Q(x)$, where the domain consists of all real numbers?

Solution:

Because $Q(x)$ is false for every real number x , the existential quantification of $Q(x)$, which is $\exists x Q(x)$, is false.



- If the elements in the universe of discourse can be listed, $U = \{x_1, x_2, \dots, x_n\}$

$$\exists x P(x) \Leftrightarrow P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$$

Example : What is the truth value of $\exists x P(x)$, where $P(x)$ is the statement " $x^2 < 10$ " and the domain consists of the positive integers not exceeding 4?

Solution : $P(x): x^2 < 10$

$U = \{\text{positive integers not exceeding 4}\}$

P(1)	P(2)	P(3)	P(4)	$P(1) \vee P(2) \vee P(3) \vee P(4)$
$1^2 < 10$ $1 < 10$ TRUE	$2^2 < 10$ $4 < 10$ TRUE	$3^2 < 10$ $9 < 10$ TRUE	$4^2 < 10$ $16 < 10$ FALSE	$T \vee T \vee T \vee F = T$

Negation of Quantifiers

- Distributing a negation operator across a quantifier changes a universal to an existential and vice versa.

$$\sim \forall x P(x) \Leftrightarrow \exists x \sim P(x)$$

$$\sim \exists x P(x) \Leftrightarrow \forall x \sim P(x)$$



Mathematical Induction

PRINCIPLE OF MATHEMATICAL INDUCTION

To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that $P(1)$ is true.

INDUCTIVE STEP: We show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all positive integers k . ie., for all positive integers k ,

if $P(k)$ is true, then $P(k + 1)$ is true.
 $P(k)$ is True $\rightarrow$ Induction Assumption

Example 1 : For any positive integer n , $1 + 2 + \dots + n = n(n+1)/2$.

Proof : by Mathematical Induction

Let's $P(n) \rightarrow 1 + 2 + \dots + n = (n(n+1))/2$.

Basis Step. We must verify that $P(1)$ is True. If $n = 1$

$$\text{L.H.S of } P(1) \rightarrow 1 \quad \text{R.H.S of } P(1) \rightarrow \frac{1(1+1)}{2} = \frac{1(2)}{2} = \frac{2}{2} = 1$$

Inductive Step: we assume that $P(k)$ is true

$$P(k) \rightarrow 1 + 2 + \dots + k = \frac{k(k+1)}{2} \quad \rightarrow \text{Induction Assumption} \quad \rightarrow \text{Eqn 1}$$

Under this assumption, it must be shown that $P(k + 1)$ is true, namely, that

$$P(k+1) \rightarrow 1 + 2 + \dots + k + (k+1) = \frac{(k+1)(k+2)}{2} \quad \text{is also True}$$

When we add $k + 1$ to both sides of the equation in $P(k)$, Eqn 1 we obtain

$$1 + 2 + \dots + k + (k+1) = \frac{k(k+1)}{2} + (k+1)$$

$$\frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}$$

From Eqn 1

$$\frac{k(k+1)}{2} + 2(k+1) = \frac{(k+1)(k+2)}{2}$$

$$k^2 + k + 2k + 2 = k^2 + 2k + k + 2$$

L.H.S = R.H.S Hence Proved



Example 2 : The sums of the first n positive odd integers for $n = 1, 2, 3, 4, 5$ are

$$1 + 3 + 5 + \dots + (2n - 1) = n^2.$$

Proof : by Mathematical Induction

$$\text{Let's } P(n) \rightarrow 1 + 3 + 5 + \dots + (2n - 1) = n^2.$$

Basis Step. We must verify that $P(1)$ is True. If $n = 1$

$$\text{LHS of } P(1) \rightarrow 1 \quad \text{RHS of } P(1) \rightarrow 1^2 = 1$$

Inductive Step: we assume that $P(k)$ is true

$$P(k) \rightarrow 1 + 3 + 5 + \dots + 2k - 1 = k^2. \quad \rightarrow \text{Induction Assumption} \quad \rightarrow \text{Eqn 1}$$

Under this assumption, it must be shown that $P(k + 1)$ is true, namely, that

$$P(k+1) \rightarrow 1 + 3 + 5 + \dots + 2k - 1 + 2(k+1) - 1 = (k+1)^2 \quad \text{is also True}$$

When we add $k + 1$ to both sides of the equation in $P(k)$, Eqn 1 we obtain

$$1 + 3 + 5 + \dots + 2k - 1 + 2(k+1) - 1 = (k+1)^2$$

$$k^2 + 2(k+1) - 1 = (k+1)^2$$

$$k^2 + 2k + 2 - 1 = (k+1)^2$$

$$k^2 + 2k + 1 = k^2 + 1 + 2k$$

L.H.S = R.H.S Hence Proved



Example 3 Prove $P(n) = n < 2^n$ for all positive integers n

Basis step: $P(1) = 1 < 2^1 = 2$ TRUE

Inductive step: assuming $P(k)$ is true, show that $(k+1) < 2^{k+1}$,
given $k < 2^k$

– Can add 1 to both sides of $P(n)$, that is:

$$k+1 < 2^k + 1$$

– Since $1 \leq 2^k$, we get:

$$k+1 < 2^k + 1 \leq 2^k + 2^k = 2^{k+1}$$